JACOB HOGUE

GEOSPATIAL ENGINEER · FULL-STACK CLOUD SYSTEMS · PYTHON AND SQL · CUSTOMER-FACING SOLUTIONS

+1 (978) 868-0097 | everythingspatialgis@gmail.com | linkedin.com/in/jacobhogue | jacob-hogue.github.io

Professional Summary

Geospatial engineer with 8+ years building full-stack production systems, from data pipelines through the interfaces people use. At Airspace Link, led 3D airspace analysis for military air bases and airports and shipped the obstruction calculator to AirHub Marketplace, and contributed to a production platform on Azure that runs **50+ federal datasets** through its pipelines for enterprise and government stakeholders. Earlier, built the City of Rochester, NH's work order system from public intake to crew dispatch and trained **100+ city staff** to run it.

Professional Experience

Airspace Link | Remote

Geospatial Engineer September 2022 – Present

Full-stack engineer on a cloud-deployed geospatial platform serving UAS stakeholders across municipal, enterprise, and federal clients.

- **Led 3D airspace analysis for military air bases and airports**, deriving FAA approach/departure surfaces and facility map grids in Python from published facility standards. Built an obstruction calculator with point and volume checks that gave immediate visual feedback on whether proposed drone flights or installations would encroach on approach, departure, or security zones. URL-parameter handoff passed operational volumes from AirHub Portal's approval workflow into the ArcGIS JavaScript SDK app deployed to AirHub Marketplace. It replaced operators' incomplete KMLs and manual Excel incursion checks. **Used by 50 senior military staff**, it **cut the time to receive a go/no-go response by 75%**.
- **Contributed to data processing on a production geospatial platform on Azure** serving thousands of UAS operators, running **50+ federal datasets** through its pipelines and maintaining those source datasets for enterprise and government stakeholders
- **Contributed to AirHub Portal from early development through production release**, building features while supporting technical proposals and RFP responses for federal three-letter agencies, municipal governments, and enterprise clients across the product's go-to-market
- **Built a browser-based drone flight simulator in Cesium (WebGL)** visualizing routes against terrain, airspace, and obstruction models. For the MHA Nation medical delivery pilot, combined HexGen risk data with LiDAR to compute a least-cost 30-mile corridor at 100ft AGL, **cutting courier time by 60%**
- **Led spatial risk assessment for the City of Romulus, MI** that elevated 34 FAA ceiling grids and expanded drone flight area by 13.92 square miles, improving UAS operational access for 5,500+ residents and ~300 businesses
- **Built a Fire Incident Response and EMS Dashboard** for the Detroit Aerial Innovation Region, analyzing 263,878 incident records across 47 fire stations comparing driving vs drone-as-first-responder response times
- **Maintain the CI/CD deployment pipeline (GitHub Actions, Docker)** with consistent releases across Dev, Stage, and Prod on Azure, ArcGIS Online, and PostgreSQL

City of Rochester, New Hampshire | Rochester, NH

GIS / Asset Management Technician March 2021 – September 2022

Technical lead for GIS infrastructure, data management, and application development serving cross-departmental city operations.

- **Built a full geospatial CMMS from backend to frontend**: public service request intake, automated supervisor routing, work order creation and crew dispatch, status dashboards (Experience Builder), and asset-linked work history for physical infrastructure. Built on SQL views, Web AppBuilder, and Collector, synced with Beehive Asset Management. **Trained 100+ city staff**, from front desk to the DPW director
- **Created a fleet management application** integrating real-time IoT sensor data from Verizon Fleet Management, enabling data-informed budgeting and cost control

- **Co-managed the city's Microsoft SQL Server Enterprise geodatabase** with **100+ feature classes** federated across ArcGIS Enterprise and Online, designing optimized SQL views and indexing strategies for cross-departmental data accessibility

---

Granite State Rural Water Association | Walpole, NH

GIS / Source Water Specialist May 2020 – March 2021

Customer-facing technical consultant providing on-site GIS training and data analysis to rural communities across New Hampshire.

- **Provided on-site technical consulting and training to 270 member water systems** across New Hampshire, delivering practical source water protection plans that non-technical stakeholders could implement independently
- **Coordinated cross-agency collaboration** with state, county, and local government partners to identify, prioritize, and implement environmental protection programs
- **Produced hydrology, geology, and soil analysis deliverables** for the Whitewater Watershed Management Plan, used directly in resource management planning

---

Antioch University New England | Keene, NH

GIS Research & Teaching Assistant August 2018 – May 2020

- **Taught Intro to GIS and Advanced GIS courses** to **~20 Master's students per class**, ranging from complete beginners to experienced professionals
- **Automated ecological co-occurrence models in Python** for Moosewood Ecological's Natural Resource Inventory for Bow and Chichester, NH, identifying Conservation Focal Areas for municipal land protection planning (Master's Capstone)
- **Built an ArcGIS Collector application** for the American Turtle Observatory to collect species abundance and habitat characteristic data for wood turtle population tracking in the field

---

New Hampshire Geological Survey | Concord, NH

Stream Crossing Assessor May 2019 – August 2019

FEMA Hazard Mitigation-funded statewide stream crossing assessment program under NH DES Hazards Program.

- **Assessed ~450 culvert and bridge stream crossings statewide** using geomorphic compatibility indices, feeding hydraulic/hydrologic models for infrastructure risk evaluation and culvert replacement prioritization
- **Processed SSURGO and LiDAR DEM data using ArcHydro tools** for watershed delineation and input into hydraulic assessment models
- **Conducted field surveys** using alidade, laser rangefinder, depth rods, and survey levels for stream cross-section and longitudinal profiling

---

Solitude Lake Management | Shrewsbury, MA

Seasonal Aquatic Technician May 2018 – August 2018

Field-based aquatic vegetation management and water quality monitoring across New England lakes and ponds for a regional aquatic and environmental services consulting firm.

- **Collected in-situ water quality data** (turbidity, pH, temperature, depth) for pre- and post-treatment monitoring of client waterbodies

---

Fitchburg State University | Fitchburg, MA

Three undergraduate research positions across planetary science, public safety analytics, and environmental hydrology (concurrent with BS degree).

Research Intern (Dr. Reid Parsons) June 2016 – August 2016

- **Mapped extent and ice-flow characteristics of Martian lobate debris aprons** as part of a joint grant with NASA Ames Research Center

Research Intern (Dr. Jane Huang) September 2016 – May 2017

- **Performed hotspot and cluster analysis mapping** of crimes, foreclosures, and building permits for the ReImagine North of Main Community Revitalization Project, collaborating with Crocker Center and Fitchburg Police Department

Senior Research Assistant, GIS (Dr. Jane Huang) May 2017 – August 2017

- **Conducted Nashua River watershed health analysis** through the Fitchburg State Student-Faculty Collaborative Summer Research Experience, integrating biological, chemical, geomorphological, and GIS perspectives

Skills

Programming and Databases: Python, TypeScript, JavaScript, SQL, R, HTML/CSS, PostgreSQL/PostGIS, Microsoft SQL Server, Supabase, SQLAlchemy

Cloud, DevOps and APIs: Azure, Google Cloud Platform (GCP), Firebase, Docker, Git, GitHub Actions, CI/CD (Continuous Integration/Continuous Deployment), REST APIs, Flask, Playwright, Postman, Production Monitoring, Technical Documentation

Geospatial and Remote Sensing: ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, ArcGIS API for Python, ArcGIS Maps SDK for JavaScript, Experience Builder, Web AppBuilder, Survey123, Field Maps, Collector, Dashboards, QGIS, GDAL, Geopandas, H3, GeoJSON, Vector Tiles, Parquet, LiDAR, ESRI Deep Learning Packages, Image Segmentation, Cesium (WebGL), Leaflet, Network Analyst, OSMnx, Valhalla, routingpy, Power Automate

Data and Visualization: ETL, Data Pipelines, Data Modeling, NumPy, Pandas, Matplotlib, D3.js, Streamlit, Jupyter, Conda, Tailwind CSS, Calcite Components, Figma, Remotion

AI Development Tooling (used daily): Claude Code (Anthropic), Claude API (Anthropic), GPT (OpenAI), Windsurf, Codex

Personal Projects

The Shape-Files November 2026 – Present

- **Write and publish The Shape-Files (jacob-hogue.github.io/the-shape-files)**, a weekly technical history of GIS told through the formats, algorithms, and systems that built it, from map rooms and tracing paper to cloud-native data and AI agents
- **Published as a static site with an RSS feed**, written for GIS analysts, developers, and data engineers who want to know why their tools behave the way they do

Datum September 2026 – Present

- **Built and released Datum (datum-gis.com)**, a public directory of **9,242 government ArcGIS servers** from **5,459 organizations** in federal, state, county, and city government, which were previously scattered with no common index
- **Automated upkeep in Python**: new servers are found through ArcGIS Hub, and every listed server gets a weekly live status check

Education

Antioch University New England | Keene, NH

Master of Science, Conservation Biology • Certificate, Applied GIS August 2018 – May 2020

Coursework: Biostatistics, Advanced GIS, Applied Spatial Analysis, Leadership for Change

Fitchburg State University | Fitchburg, MA

Bachelor of Science, Environmental & Earth Sciences • Summa Cum Laude, 3.99 GPA August 2014 – May 2018

Minor: Geographic Sciences & Technology • President's List all semesters

Coursework: Computer Science, GIS, Remote Sensing of the Environment, Computer Cartography, Earth Systems Science